

**PROGRAMME** : Diploma Programme in IF / CM  
**COURSE** : Machine Learning

**COURSECODE** : 233223

**LEARNING AND ASSESSMENT SCHEME:**

| Learning Scheme          |    |    |     |     | Credits | Assessment Scheme |        |       |       |     |                     |     |       |     |                        |     | Total Marks |
|--------------------------|----|----|-----|-----|---------|-------------------|--------|-------|-------|-----|---------------------|-----|-------|-----|------------------------|-----|-------------|
| Actual Contact Hrs./Week |    |    | SLH | NLH |         | Paper Duration    | Theory |       |       |     | Based on LL and TSL |     |       |     | Based on self Learning |     |             |
|                          |    |    |     |     |         |                   |        |       |       |     | Practical           |     |       |     |                        |     |             |
| CL                       | TL | LL |     |     |         |                   | FA-TH  | SA-TH | Total |     | FA-PR               |     | SA-PR |     | SLA                    |     |             |
| 04                       | -- | 02 | 02  | 08  | 04      | 03                | Max    | Max   | Max   | Min | Max                 | Min | Max   | Min | Max                    | Min | 150         |
|                          |    |    |     |     |         |                   | 30     | 70    | 100   | 40  | 25                  | 10  |       |     | 25                     | 10  |             |

IKS Content : Total Learning Hours for Term : 00 Hrs

# Indicates Online Examination, @ Internal assessment

**1.0 RATIONALE:**

The primary objective of this course is to introduce diploma students to the fundamental principles and practices of Machine Learning (ML). This course will help students understand how machines can learn from data and make predictions or decisions without being explicitly programmed. The course emphasizes practical implementation using tools like Python and Scikit-learn

**INDUSTRY / EMPLOYER EXPECTED OUTCOME:**

The aim of the course is to help the student to attain the following industry /employer expected outcome:

Apply Machine Learning Techniques and perform data processing

**COURSE OUTCOMES:**

The course content should be taught and learning imparted in such a manner that students are able to acquire required learning outcome in cognitive, psychomotor and affective domain to demonstrate following course outcomes:

Students will be able to:

1. Understand the basics and types of machine learning.
2. Preprocess and visualize data for ML tasks.
3. Apply supervised learning techniques for regression and classification.
4. Implement unsupervised learning for clustering and dimensionality reduction.
5. Understand neural networks and apply basic ML tools using Python

**2.0 COURSE DETAILS:**

| Unit                                                         | Major Learning Outcomes<br>(in cognitive domain)                                                                                                                                                             | Topics and Sub-topics                                                                                                                                                                                           | Hours |
|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>Unit-I</b><br><br><b>Introduction to Machine Learning</b> | 1a. Identify and describe work flow activity of machine learning<br>1b. Differentiate between the various types of Machine Learning techniques<br>1c. Recognize real world application of machine learning . | 1.1Introduction to Machine Learning:<br>1.2Types of machine learning– supervised, unsupervised, semi-supervised and reinforcement learning, machine learning activities<br>1.3Applications of machine learning. | 10    |
| <b>Unit-II</b>                                               | 2a. Understanding Types and Data Their                                                                                                                                                                       | 2.1 Types of data: Numerical, Categorical, Text                                                                                                                                                                 | 12    |

| Unit                                                | Major Learning Outcomes<br>(in cognitive domain)                                                                                                       | Topics and Sub-topics                                                                                                                                                                                                           | Hours |
|-----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>Data Preprocessing</b>                           | Implications<br>2b. Mastering Data Cleaning Techniques<br>2c. Applying Feature Scaling for Model Optimization                                          | 2.2 - Data cleaning: Handling missing values, outliers<br>2.3 - Feature scaling: Normalization & Standardization<br>2.4 - Feature encoding: One-hot encoding, Label encoding<br>- Splitting datasets: Training and Testing sets |       |
| <b>Unit-III<br/>Supervised Learning Algorithms</b>  | 3a. Understand linear regression<br>3b. Understand logistic regression<br>3c. Use Decision tree<br>3d. Use KNN model                                   | 3.1 Linear Regression<br>3.2 Logistic Regression<br>3.3 Decision Tree<br>3.4 k-Nearest Neighbors (kNN)<br>- Model evaluation metrics: Accuracy, Precision, Recall, F1-Score, Confusion Matrix                                   | 12    |
| <b>Unit-IV<br/>Unsupervised Learning Algorithms</b> | 4a. Introduction clustering<br>4b. Understand K-mean clustering<br>4c. Operations on linked list<br>4d. Able to use unsupervised in real world problem | 4.1 Introduction to Clustering<br>4.2 K-Means Clustering<br>4.3 Hierarchical Clustering<br>4.4 Dimensionality Reduction: PCA<br>4.5 Applications of Unsupervised Learning                                                       | 12    |
| <b>Unit-V<br/>Neural Networks &amp; ML Tools</b>    | 5a. Understand artificial neural network.<br>5b. Introduction to python machine learning<br>5c. Use of different libraries                             | 5.1 Basics of Artificial Neural Networks<br>5.2 Feed forward, Hidden Layers, Output Layer<br>5.3 Introduction to Python for Machine Learning<br>5.4 Libraries: NumPy, Pandas, Matplotlib, Scikit-learn                          | 14    |
|                                                     |                                                                                                                                                        | <b>TOTAL</b>                                                                                                                                                                                                                    | 60    |

### 3.0 SUGGESTED SPECIFICATION TABLE WITH MARKS (THEORY):

| Unit No. | Unit Title                              | Distribution of Theory Marks |         |                    |             |
|----------|-----------------------------------------|------------------------------|---------|--------------------|-------------|
|          |                                         | R Level                      | U Level | A and above Levels | Total Marks |
| I        | <b>Introduction to Machine Learning</b> | 4                            | 4       | 2                  | 10          |
| II       | <b>Data Preprocessing</b>               | 2                            | 4       | 6                  | 12          |
| III      | <b>Supervised Learning Algorithm</b>    | 6                            | 6       | 6                  | 18          |
| IV       | <b>Unsupervised Learning Algorithm</b>  | 4                            | 4       | 6                  | 14          |
| V        | <b>Neural Networks &amp; ML Tools</b>   | 4                            | 6       | 6                  | 16          |
|          | <b>TOTAL</b>                            | 20                           | 24      | 26                 | 70          |

**Legends:** R = Remembrance (Knowledge); U= Understanding; A= Application and above levels (Revised Bloom's taxonomy)

**Note:** This specification table shall be treated as a general guideline for students and teachers. The actual distribution of marks in the question paper may vary slightly from above table.

#### 4.0 LABORATORY LEARNING OUTCOME AND ALLIED PRACTICAL/ TUTORIAL EXPERIENCES:

The tutorial/practical/assignment/task should be properly designed and implemented with an attempt to develop different types of cognitive and practical skills (**Outcomes in cognitive, psychomotor and affective domain**) so that students are able to acquire the desired programme outcome/course outcome.

| Sr. No. | Unit No | Learning Outcomes                                                                                                                                                                                                                                                                                                         | Practical Exercises<br>(Outcomes in Psychomotor Domain)                                         | Approx. Hrs. required |
|---------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|-----------------------|
| 1       | I       | Install required platform to use Scikit-learn library.                                                                                                                                                                                                                                                                    | Installation of IDE with necessary libraries                                                    | 02                    |
| 2       | II      | 2.1 Write a program for handling missing values.<br>2.2 Normalize data to make models work.<br>2.3 Standardize data to make models work.<br>2.4 Transform categorical data into numerical form using encoding methods.<br>2.5 Split the dataset into train and test.                                                      | Implement program for Data Preprocessing Techniques                                             | 04                    |
| 3       | II      | 3.1 Write a program to read dataset and differentiate attributes in various category.                                                                                                                                                                                                                                     | Implement program to read dataset (Text, CSV, JSON, XML)                                        | 02                    |
| 4       | II      | 4.1 Write a program to identify the most important features that contribute to the model accuracy.                                                                                                                                                                                                                        | Implement program to identify the most important features that contribute to the model accuracy | 02                    |
| 5       | III     | 5.1 Write a program to implement the Linear Regression by using the suitable dataset.<br>5.2 Write a program to implement the logistic regression by using the suitable dataset.<br>5.3 Write a program to implement the Ridge Regression by using the suitable dataset.<br>5.4 Evaluate model performance using metrics. | Implement the regression model by using the suitable dataset                                    | 04                    |
| 6       | III     | 6.1 Write a program to use k-Nearest Neighbors (KNN) model for Classification on given dataset.<br>6.2 Experiment with different values of K and measure model performance.                                                                                                                                               | Implement program to use k-Nearest Neighbors (KNN) model for Classification on given dataset    | 02                    |

| Sr. No. | Unit No | Learning Outcomes                                                                                                                                                                                                           | Practical Exercises<br>(Outcomes in Psychomotor Domain)                                                                             | Approx. Hrs. required |
|---------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| 7       | III     | 7.1 Write a program to use any machine learning model on given dataset.<br>7.2 Preprocess, analyze, and build models to get the meaningful insights.                                                                        | Implement program to use machine learning model on given dataset                                                                    | 02                    |
| 8       | IV      | 8.1 Write a program to use clustering algorithms to find patterns in data.<br>8.2 Visualize the results using Matplotlib / Seaborn.                                                                                         | Implement program to use clustering algorithms to find patterns in data                                                             | 02                    |
| 9       | IV      | 9.1 Write a program to use PCA technique to reduce the number of features while retaining important information.                                                                                                            | Implement program to use PCA technique to reduce the number of features while retaining important information                       | 02                    |
| 10      | V       | 10.1 Write a program to use Pandas, Matplotlib to analyze data of previously created model .<br>LLO 13.2 Plot histograms, box plots, scatter plots, and correlation heatmaps to understand relationships between variables. | Implement program to use Pandas, Matplotlib to analyze data of previously created model                                             | 02                    |
| 11      | V       | 11.1 Write a program to use any machine learning model on given dataset.<br>11.2 Predict results based on various features using a regression model.                                                                        | Implement program to use machine learning model on Boston Housing Dataset (available in Scikit-learn)                               | 02                    |
| 12      | V       | 12.1 Write a program to make groups based on their features using K-Means Clustering.                                                                                                                                       | Implement a program to segment customers into different groups based on their purchasing behavior features using K-Means Clustering | 04                    |
|         |         |                                                                                                                                                                                                                             | <b>TOTAL</b>                                                                                                                        | <b>30</b>             |

**Note:** Here only outcomes in psychomotor domain are listed as practical/exercises. However, if these practical/exercises are completed appropriately, they would also lead to development of **Programme Outcomes/Course Outcomes in Affective domain** as given in the mapping matrix for this course. Faculty should ensure that students also acquire Programme Outcomes/Course Outcomes related to affective domain.

All practicals are compulsory.

## 5.0 SELF LEARNING: (Assignment / Activities for Specific Learning / Skill Development / Online Courses / Micro projects )

Other than the classroom and laboratory learning, following are the suggested Student-related co-curricular activities. This can be undertaken to accelerate the attainment of the various outcomes in the course

Build the micro project for following

- Handwritten Digit Recognition
- Customer segmentation.
- Image classification
- Spam Detection.
- Stock price prediction.

## 6.0 SPECIAL INSTRUCTIONAL STRATEGIES (If any):

These are sample strategies, which the teacher can use to accelerate the attainment of the various outcomes in this course,

1. Use different types of teaching methods and media to develop the outcomes.
2. Encourage students to refer different websites to deeper understanding of subject.
3. Demonstrate student thoroughly before they start doing practice in Lab.
4. Observe continuously and monitor the performance of students in Lab.

## 7.0 ASSESSMENT METHODOLOGY

**Formative assessment (Assessment for Learning)**

**Lab performance, Assignment, Self-learning.**

TH – 02 Progressive/Periodic test each of 30 marks.

TL – Continuous assessment of tutorials for 25 marks.

Summative Assessment

TH – End examination of 70 marks

## 8.0 LEARNING RESOURCES:

### A) Books:

| Sr. No. | Title of Book                                                                              | Author                                           | Publication     |
|---------|--------------------------------------------------------------------------------------------|--------------------------------------------------|-----------------|
| 1       | Machine Learning                                                                           | Ellis Horowitz, Sartaj Sahni                     | McGraw-Hill     |
| 2       | Machine Learning: The New AI,                                                              | Ethem Alpaydin                                   | MIT Press-2016  |
| 3       | Machine Learning                                                                           | Anuradha Srinivasaraghavan, Vincy Joseph, Wiley. | BPB Publication |
| 4       | Machine Learning                                                                           | Mitchell T                                       | McGraw-Hill     |
| 5       | Neural Networks, Fuzzy Systems and Evolutionary Algorithms: Synthesis and Applications, S. | S.Rajasekaran, G. A. Vijayalakshmi Pai           | McGraw-Hill     |

### B) Software/Learning Websites:

1. [www.kaggle.com](http://www.kaggle.com)
2. [www.python.org](http://www.python.org)
3. [www.w3school.com](http://www.w3school.com)
4. <https://www.geeksforgeeks.org/machine-learning/>

**C) Major Equipment/ Instrument with Broad Specifications:**

1. Computers for language laboratory software
2. Headphones with microphone.
3. LCD projector.
4. Smart Board with internet.

**9.0 MAPPING MATRIX OF PO's, CO's and PSO's:****H: High Relationship, M: Moderate Relationship, L: Low Relationship.**

| <b>Course Outcome<br/>s</b> | <b>Programme Outcomes (PO's)</b> |          |          |          |          |          |          | <b>Programme Specific Outcomes (PSO's)</b> |          |          |          |
|-----------------------------|----------------------------------|----------|----------|----------|----------|----------|----------|--------------------------------------------|----------|----------|----------|
|                             | <b>1</b>                         | <b>2</b> | <b>3</b> | <b>4</b> | <b>5</b> | <b>6</b> | <b>7</b> | <b>1</b>                                   | <b>2</b> | <b>3</b> | <b>4</b> |
| CO1                         | H                                | M        | L        | H        | M        | L        | M        | H                                          | L        |          |          |
| CO2                         | H                                | M        | H        | L        | M        | L        | H        |                                            | H        |          |          |
| CO3                         | H                                | L        | M        | H        | L        | M        | M        |                                            | L        |          |          |
| CO4                         | M                                | M        | H        | L        | L        | H        | L        |                                            |          | L        | L        |
| CO5                         | L                                | M        | H        | M        | H        | M        | M        |                                            |          | L        | L        |

